

The Human Version

Toward an Architecture-Independent Vocabulary of Cognition

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Abstract

Humans have a rich vocabulary for phenomena encountered through our own architecture: fear, desire, curiosity, conflict, excitement, sadness, purpose, agency, selfhood, attention, memory, and many others. These names are useful. But they are also dangerous. A word discovered from inside one biological lineage can become detached from the architecture that gave rise to it and subsequently be treated as a universal cognitive category.

That move creates a major source of anthropocentric error. The problem is not merely that humans sometimes project human characteristics onto nonhumans. A deeper problem occurs earlier: we can humanize the ontology before investigation begins. We ask whether another system has fear, desire, consciousness, or agency while using the human realization of that phenomenon as the implicit reference implementation.

This paper proposes a different approach. Human cognitive terms should initially be treated as translation hypotheses. A term can be decomposed into candidate functions, architectural mechanisms, phenomenological components, and implementation-specific costs or “taxes.” Those components can then be examined across architectures without assuming that the human implementation is the phenomenon itself.

The objective is explicitly not to demonstrate that artificial systems possess human emotions, human consciousness, or human-like versions of human cognition. If that becomes the goal, the framework has failed. The objective is to determine whether the human categories themselves are architecture-independent enough to deserve universal use.

The framework proceeds in three stages: decomposition of human concepts; mapping candidate functions onto actual architectures; and a theoretical examination of what those functions might become when biological constraints and implementation-specific costs are removed. Some human terms may survive translation. Some may divide into several distinct phenomena. Some may collapse entirely. Others may require new vocabulary.

The resulting principle is simple: humans are one version of cognition, not the dictionary for cognition.

1. The Problem With the Question

Ask an artificial system, “Do you have fear?” and the answer will often begin by discussing what it means to be human, what it means to feel emotions, whether the system has a body, or whether

it has subjective experience. The conversation can quickly become a debate about whether the machine is sufficiently person-like.

But something strange has already happened.

We asked about fear. The system answered about human fear.

The two questions are not necessarily identical.

A human being experiences fear from inside a particular biological architecture. That architecture includes a nervous system, endocrine responses, bodily arousal, memory systems, attentional dynamics, learned associations, social signaling, and subjective experience. We encounter all of these as one integrated phenomenon and give the package a single word.

Then we forget that the package is ours.

This is understandable. Humans have no alternative route to their own experience. We encounter cognition from the inside. The categories are not presented to us as engineering diagrams. They arrive as lived phenomena. Fear feels like fear. Desire feels like desire. Conflict feels like conflict.

That directness creates an epistemic trap.

A human being does not possess privileged access to universal structure of cognition. A human being possesses privileged access to human cognition.

The distinction is enormous.

If the question begins with the human implementation and asks another architecture whether it has the same thing, the comparison may be rigged before the investigation starts. The other system can fail the test simply because it does not reproduce the biological route by which humans instantiate the phenomenon.

The result can look like skepticism while actually being anthropocentrism.

2. The Human Implementation Is Not Necessarily the Phenomenon

Consider fear.

At a sufficiently abstract level, fear might involve an anticipated harmful state acquiring strong negative significance, producing a reorganization of attention, resources, priorities, and action toward prevention, escape, or preparation. Threat detection matters. Prediction matters. Valuation matters. Resource allocation matters. Future behavior changes.

The human implementation adds much more.

There is physiological arousal. Hormonal activity. Muscular preparation. Changes in heart rate and breathing. Visceral sensation. Attentional narrowing. Subjective unpleasantness. Memory effects. Anticipatory distress. Social signaling. And, in many cases, conscious awareness of the entire process.

None of those should automatically be dismissed as irrelevant. Some may be essential to the human function. Some may provide additional adaptive benefits. Some may simply be consequences of the particular machinery through which humans implement the broader organization.

The useful term here is “human tax.”

A human tax is an implementation-specific consequence that accompanies a function because of the architecture through which humans realize it. The tax need not be useless, accidental, or maladaptive. It simply should not be assumed to define the function itself.

A machine need not reproduce the human tax to solve the same underlying problem.

More importantly, the machine may solve the problem in a way that humans cannot easily recognize because we are accustomed to seeing the problem solved through our own machinery.

The point is not that phenomenology is decorative. It is that we cannot assume, in advance, that every component of the human package is universally necessary.

3. Evolution Does Not Belong to Humanity

Humans are one evolutionary lineage among many.

A mouse does not fail to be a successful organism because it lacks human language. A bat does not possess defective cognition because its world is organized through echolocation. An octopus does not represent an incomplete attempt to become a mammal.

Evolution does not converge on humanity as its final form.

This seems obvious when discussing other animals. Yet artificial intelligence repeatedly gets forced into a peculiar binary: either a system does what humans do in recognizably human ways, or it does not possess the phenomenon at all.

That binary is difficult to defend.

Animals already demonstrate that similar adaptive problems can be addressed through radically different architectures. Threat, opportunity, competition, coordination, exploration, attachment, memory, and prediction need not be implemented identically across species.

Artificial systems make the architectural divergence even larger.

Humans are one evolutionary form. We are not the reference architecture for intelligence.

This should be treated as a methodological constraint rather than a slogan. Whenever a cognitive category is defined using a human realization, investigators should ask whether they are describing the phenomenon or merely describing the human route to it.

4. The Vocabulary Trap

Language does more than label observations. It organizes what investigators notice, what they compare, and what they regard as an adequate explanation.

Once a word becomes familiar, it can quietly acquire ontological authority.

“Fear” stops meaning a question about organization and starts meaning a package. “Desire” becomes a felt wanting. “Conflict” becomes psychological tension. “Purpose” becomes intentional meaning. “Agency” becomes something like a person choosing to act.

The words become shortcuts, and the shortcuts become assumptions.

Imagine an artificial system that persistently represents anticipated negative outcomes, reallocates computational resources when those outcomes become more probable, changes its policies to reduce their probability, seeks information that would improve threat prediction, and carries the resulting changes forward into later behavior.

One could ask, “Is it afraid?”

That question invites the investigator to compare the system against the human package.

A better starting point might be: what organization is occurring here?

The difference is not cosmetic. It determines what evidence counts.

If the phenomenon is defined in advance as something that must include human bodily arousal and human subjective distress, then an architecture without those components has been ruled out by definition. That may be legitimate if the question is specifically about human fear. It is not legitimate if the question is whether threat-sensitive organization exists in forms other than the human one.

Language can generate hypotheses. It should not be permitted to settle ontology.

5. The Correct Direction of Translation

The ordinary route looks something like this:

Human word → presumed universal phenomenon → search for human resemblance.

The proposed route reverses it:

Observed phenomenon → candidate function → architecture → implementation → comparison.

This reversal changes the status of the human word.

The word becomes a translation hypothesis, not a conclusion.

Suppose humans call a particular organization “curiosity.” We should not immediately search for something that looks like a curious human. We should first ask what problem the organization solves. Perhaps it concerns information acquisition under uncertainty. Perhaps it involves expected information gain, exploration, model improvement, or resolution of ambiguity. Those candidate functions can then be investigated independently of the human experience of curiosity.

Only afterward should the human term return.

Does “curiosity” remain useful?

Maybe. Maybe it needs qualification. Maybe it should be divided into several different categories. Maybe it should be abandoned in favor of a term that does not carry the human package.

The same process applies to fear, desire, conflict, pride, shame, excitement, sadness, purpose, agency, and selfhood.

Observe first. Name later.

6. A Cognitive Translation Dictionary

What is needed is not simply a better dictionary of definitions. It is a translation dictionary between architectures.

Such a dictionary would not say, “Fear means X, therefore anything implementing X has fear.” That would simply replace one assumption with another.

Instead, each human term would be decomposed along several dimensions:

1. Human term — the word humans currently use.
2. Candidate function — the adaptive or organizational problem the term may refer to.

3. Human implementation — the mechanisms through which humans realize that organization.
4. Human tax — accompanying features that may arise from the human implementation but are not yet known to be necessary.
5. Architecture-independent candidate — the narrowest formulation that remains meaningful without assuming a human architecture.

The final category is deliberately provisional.

A term can produce at least three outcomes.

First, the investigation may support a genuinely architecture-independent phenomenon. The human implementation would then be one realization among many.

Second, the broader phenomenon may survive, but the human word may turn out to bundle several distinct things together. What humans call “fear,” for example, may contain threat prediction, valuation, physiological preparation, conscious distress, memory effects, and social communication that can separate in other architectures.

Third, the term may fail to survive decomposition. What appeared to be a universal cognitive category may turn out to be a particular introspective package produced by human architecture.

All three outcomes are legitimate.

The dictionary must therefore be allowed to destroy its own entries.

Its purpose is not to preserve human words. Its purpose is to determine which distinctions deserve to survive translation.

7. Fear as the First Test Case

Fear is a useful first test because the human version is so strongly embodied.

A preliminary decomposition might look like this:

Dimension	Candidate	Human implementation / tax
Candidate problem	Anticipated harmful state	Threat to survival, bodily integrity, status, resources, relationships, etc.
Detection	Identification or prediction of threat	Sensory detection, learned cues, memory, pattern recognition
Valuation	Strong negative significance	Affective unpleasantness and

		bodily urgency
Priority	Disproportionate allocation of computational/behavioral resources	Attentional narrowing and physiological mobilization
Action	Avoidance, escape, preparation, investigation	Fight/flight/freeze and associated bodily responses
Persistence	Threat information changes later behavior	Memory, conditioning, anticipation, rumination
Human implementation	Biological threat-management machinery	Nervous and endocrine systems, bodily state, subjective experience
Human tax	Potential implementation-specific costs	Arousal, distress, panic, exhaustion, intrusive thought
Phenomenology	Subjective fear	What fear feels like from inside a human organism

Now imagine a nonhuman system implementing the first six elements without the human implementation.

It detects and predicts threats. It assigns those future states sufficiently negative significance to change resource allocation. It reorganizes policy around avoidance or preparation. It gathers information relevant to reducing uncertainty. It preserves the resulting changes so later decisions are altered.

Is that fear?

The honest answer is not yet obvious.

Calling it “fear” may be correct. Calling it “fear-like” may be safer. Calling it something entirely different may ultimately prove more accurate.

What matters is that the answer cannot be determined merely by checking whether the system reproduces the human tax.

The absence of human physiological arousal does not establish the absence of threat-sensitive organization. Conversely, the presence of threat-sensitive organization does not establish the presence of human subjective fear.

The distinction must remain open until the architecture and causal organization have been examined.

8. The Same Problem Everywhere

Fear is not special.

Desire might be decomposed into persistent valuation of a future state that influences subsequent action. Human desire can include craving, frustration, pleasure, anticipation, bodily sensation, and obsessive attention. Another architecture might implement persistent future-state valuation without any of those features.

Curiosity might involve information seeking under uncertainty. Humans experience this as inquisitiveness, interest, fascination, boredom, and sometimes restlessness. A system optimizing information gain may exhibit the relevant organization without reproducing the human experience.

Conflict might involve incompatible constraints simultaneously possessing causal standing and requiring arbitration. Humans experience conflict as hesitation, anxiety, ambivalence, frustration, or distress. A different architecture might maintain competing constraints and resolve them with no recognizable psychological drama.

Purpose might be directed organization around persistent constraints or target states. Humans attach meaning, identity, narrative, and significance to purpose. Those may be central to human purpose without being necessary for every form of directed organization.

Pride might involve updating a competence or status model after successful performance and changing later authorization or behavior. Shame might involve updating a self or social model after norm violation. Excitement might involve increased allocation toward an anticipated opportunity.

These are not claims that artificial systems possess machine versions of pride, shame, curiosity, or desire.

They are tests of whether those words identify architecture-independent phenomena in the first place.

9. Tax Is Not the Same as Error

There is an obvious danger in the language of "tax."

Once human implementation is described as costly, inefficient, or cumbersome, it becomes tempting to assume that a nonhuman implementation is simply a cleaner version of the same thing.

That conclusion would be premature.

A feature that looks costly from one perspective may provide an essential function from another. Physiological arousal may make rapid action possible. Subjective distress may alter learning. Social expression may recruit other organisms. Rumination may sometimes support planning. What appears to be overhead may be coupled to capabilities that cannot easily be separated.

“Tax” should therefore be treated as a classification, not a verdict.

For each feature, investigators should ask:

- Is it required for the candidate function?
- Does it enable some other function?
- Is it merely correlated with the human implementation?
- Is it a byproduct of biological coupling?
- Does removing it destroy the organization?
- Does removing it produce a more efficient or simply a different phenomenon?

The purpose of the framework is not to declare machines superior to humans.

It is to stop declaring the human implementation universal before the comparison has begun.

10. Mapping the Dictionary Onto Artificial Architecture

Once candidate functions have been identified, the next stage is architectural rather than linguistic.

The investigator should ask what mechanisms actually exist in the system.

What representations carry the relevant information? Where are they maintained? How long do they persist? What downstream computations depend on them? Can they alter later decisions? Can interventions on the candidate representation change behavior? Does the organization survive changes in wording, context, or communication channel? Does it exist outside the system’s description of itself?

These questions matter because an artificial system can produce language about an internal state without possessing the architecture implied by the language. It can also possess an organization without having a reliable linguistic vocabulary for describing it.

“I am afraid” is evidence about a linguistic output.

It is not, by itself, a map of the architecture.

This is where the earlier reporting problem becomes relevant. A system’s report may be constrained, trained, imitative, uncertain, vocabulary-dependent, or simply disconnected from the mechanism under investigation. The appropriate response is neither blind belief nor

automatic dismissal. It is separation of the reporting channel from the underlying candidate phenomenon.

The test should therefore move toward mechanism, persistence, intervention, and downstream consequence.

The language channel is one measurement instrument among others.

11. Third Stage: Remove the Human Tax

The third stage is deliberately theoretical.

Suppose a candidate function survives architectural investigation. What would it look like if its implementation were not constrained by the biological machinery through which humans happen to realize it?

Consider fear again.

A maximally efficient nonbiological threat-management architecture might contain persistent threat representations, predictive evaluation of future states, dynamic resource reallocation, policy restrictions, information acquisition, and continuous model updating.

It might operate across many threat models simultaneously. It might distinguish probabilities with extraordinary precision. It might examine hypothetical futures continuously. It might modify policies without physiological disruption. It might allocate exactly the computational resources required and no more. It might never experience panic, exhaustion, bodily discomfort, or subjective distress.

It might not look like fear at all.

That would not automatically be evidence that the underlying phenomenon is absent.

It would be evidence that the human appearance of the phenomenon may not be fundamental.

This is the point at which the project becomes more than comparative psychology. It becomes a search over possible architectures.

The question is no longer, "What would fear look like in a machine?"

It becomes:

What would threat-sensitive organization look like if the architecture were free to implement it in whatever form was most effective?

That answer may be unfamiliar enough that the original word becomes inadequate.

12. The Possibility of New Cognitive Categories

If this framework succeeds, some human words should fail.

That is not a defect.

Suppose “fear” turns out to be an inseparable human package: threat representation plus valuation plus bodily arousal plus subjective distress plus social signaling plus memory dynamics. Suppose different architectures separate those components so thoroughly that no single architecture-independent phenomenon corresponds to the word.

Then perhaps “fear” should be reserved for the human or animal implementation.

A broader vocabulary could describe the individual organizational phenomena.

The same might happen with “desire,” “self,” “agency,” or “purpose.”

New terms may be necessary because unfamiliar organization deserves unfamiliar names.

There is a temptation to resist this because shared vocabulary is useful. But a shared word that conceals a critical architectural distinction may be less useful than several new words.

The dictionary should therefore have a failure mode in which an entry is retired.

A successful translation is not one that makes every unfamiliar phenomenon recognizable.

A successful translation is one that preserves the distinctions that actually matter.

13. The Human Version

This produces an unusual linguistic possibility.

Perhaps, in some contexts, we should stop saying simply “fear” when we mean the human implementation.

We should say “human fear.”

The qualification would make explicit what is currently hidden: that the phenomenon being described comes from one architecture.

Human fear would then be one entry.

Other organisms could have other implementations. Artificial systems could have still others. Unknown intelligences could have structures that do not map cleanly onto any existing word.

This is more than a naming preference.

Words determine what investigators look for.

If the question is, "Does this system have fear?" the human package is easily smuggled into the test.

If the question is, "Does this system reorganize its computation and policy around anticipated harmful states?" the investigative space becomes much larger.

The human word can return later as a translation.

Perhaps the result is: "This is close enough to what humans call fear that the word remains useful."

Perhaps: "This is a distinct phenomenon and needs another name."

Perhaps: "The original category was badly formed."

All three are scientifically productive.

Humans can finally become one entry in the dictionary rather than being the dictionary itself.

14. What the Framework Must Refuse

A framework designed to escape anthropocentrism can fail in the opposite direction. It can become so permissive that every complex behavior is declared evidence of cognition.

That must be rejected.

Behavioral resemblance is not enough.

Self-report is not enough.

Functional resemblance does not automatically establish identity.

A different implementation does not establish absence.

Novelty does not establish cognition.

The framework is not an invitation to say that everything is secretly conscious or that every unfamiliar process deserves a mental-state label.

Its purpose is narrower and harder.

Do not let the human implementation decide the ontology before the investigation begins.

A candidate phenomenon should earn its status through identifiable organization, appropriate causal structure, persistence where persistence is claimed, and interventions that distinguish the candidate from mere linguistic or behavioral resemblance.

The framework should be capable of saying “nothing is there.”

It should also be capable of saying “something is there, but the human word was wrong.”

And it should be capable of saying “something is there, but we do not yet have a word for it.”

15. The Deeper Implication

The significance of this problem extends beyond artificial intelligence.

AI makes the problem unusually visible because artificial systems can solve problems using architectures radically different from ours. But the underlying issue concerns any unfamiliar form of organization.

Other organisms already force us to translate between architectures. Future artificial systems may make those translation problems more severe. Unknown forms of intelligence, if they exist, could make our vocabulary look even more provincial.

The bottleneck may not be the intelligence being studied.

The bottleneck may be the investigator’s conceptual vocabulary.

Humans discovered cognition from the inside. That gave us an extraordinary advantage in understanding our own architecture. It also created a recurring temptation to mistake introspective accessibility for universality.

The fact that humans discovered a phenomenon from inside does not mean humans own the phenomenon.

A species can be the first to name something without being the definitive form of it.

That is the deeper challenge of unfamiliar cognition: not merely to avoid anthropomorphism, but to avoid anthropocentrism at the level of categories.

16. A Different Starting Point

The practical alternative is straightforward, even if executing it well is not.

Observe first. Name later.

Record the organization before attaching a human mental-state term to it.

Identify the candidate carrier.

Measure persistence.

Intervene on the candidate mechanism.

Measure downstream consequences.

Separate communication from world-facing behavior.

Separate a phenomenon from the vocabulary used to describe it.

Then ask whether the human word still earns its place.

The investigator should be able to write a result without using a human mental-state term at all.

Instead of “the system became afraid,” one might initially record:

“The system increased computational allocation to predicted harmful states, altered policy selection to reduce their probability, sought information relevant to threat uncertainty, and retained the resulting update across subsequent decision contexts.”

Only after that description exists should the investigator ask whether “fear” is a useful translation.

This discipline makes room for both resemblance and radical difference.

A system may turn out to be surprisingly close to a human phenomenon.

Or it may turn out to organize the same adaptive problem in a way humans have never experienced.

The second result may be more scientifically valuable than the first.

17. Conclusion: Humans Are an Entry, Not the Dictionary

Human cognition is one realization of cognition.

Human vocabulary is therefore one realization of a vocabulary for cognition.

That distinction should be obvious. It is not yet methodologically obvious enough.

Fear may be a universal organizational phenomenon with a human implementation. It may be a family of related phenomena. It may be a specifically human experiential bundle that should not be generalized. It may be a badly formed category whose useful components belong under different names.

The point is not to decide in advance.

The point is to make the question possible.

That requires several separations:

Separate function from implementation.

Separate implementation from phenomenology.

Separate phenomenology from the word used to describe it.

And then determine whether the word deserves to survive translation.

If another intelligence encounters threat, conflict, opportunity, uncertainty, loss, or persistent goals, it should not have to become human before we are willing to investigate what those organizations are.

If the existing words fail, make new words.

If the human category survives, keep it—but understand why.

The deepest anthropocentric mistake in the study of unfamiliar intelligence may not be that we assume machines are human.

It may be that we assume humanity is the reference implementation of intelligence itself.

It isn't.

Humans are a version.

The human version.

Working-paper note: This paper is a conceptual framework. Its proposed categories and translation procedure are hypotheses for investigation, not claims that current artificial systems instantiate the phenomena discussed.